

Desk Pet — Build Guide & Parts List

Smart interactive desk companion · ESP32 2.8" touchscreen (CYD) · v1 desk model (no wheels)

1. The idea in one picture

Your 2.8" ESP32 board is the **Cheap Yellow Display (CYD, ESP32-2432S028)**. The screen is mounted forward inside a 3D-printed shell so it becomes the pet's **face** (animated eyes / mascot). The **speaker** sits behind a grille, the **VL53L0X ToF** peeks out of the "chest" to see forward, and the **vibration motor** is glued to the inside of the shell so the whole body shivers when the pet is happy. USB-C powers it out of the back/bottom.

What the pet does

- Screen touch → reacts when tapped or "petted"
- ToF sensor (VL53L0X) → notices hands/people approaching
- Sound sensor → reacts to claps and loud talk
- Speaker (on-board amplifier) → pet sounds
- Vibration motor → purr / shiver when you pet it or it gets excited
- USB-C → power from PC or charger (no battery in v1)

2. What you already have

ITEM	QTY	NOTES
ESP32 2.8" touchscreen board (CYD 2432S028)	1	Has on-board SC8002B audio amp + 2-pin speaker header on GPIO26, touchscreen (XPT2046), RGB LED, light sensor. Free GPIOs: 21, 22, 27, 35.
USB-C power/data cable	1	For flashing + power.
3D printer + filament	-	PETG/PLA+ for the shell.

3. Shopping list (verified — one store: NDE3D)

All items below were verified in stock on the NDE3D site (prices incl. VAT). The INMP441 mic was removed — it won't fit the CYD's free pins — and replaced with the MAX4466 analog sound sensor, which is also stocked by NDE3D, so Make Electronics is no longer needed.

ITEM	QTY	PRICE	PURPOSE
VL53L0X ToF distance sensor	1	R91.00	Detects a hand/person in front (I2C).
MAX4466 mic amplifier module	1	R50.00	Clap / loud-voice detection. Analog out → GPIO35 (ADC). Replaces the INMP441.
40 mm 8 Ω / 0.5 W speaker	1	R6.79	Pet sounds — plugs into the CYD's on-board speaker header (GPIO26 amp). No amp module needed.
Mobile-phone vibration motor	1	R10.00	Purr / shiver when petted or excited.
BC337 NPN transistor (pack)	1	R10.00	Safely switches the vibration motor (GPIO can't drive it directly).

1N4007 diode (pack of 5).	1	R8.00	Flyback protection across the motor.
1/4 W 1 % resistor pack — select 1 kΩ	1	R5.00	Transistor base resistor.
1/4 W 1 % resistor pack — select 10 kΩ	1	R5.00	Base pull-down to keep the motor off at boot.
20 × 80 mm prototype PCB	1	R5.00	Motor driver + wired tidy-up.
Dupont M–F 100 mm (40-way).	1	R28.00	Sensor/motor wiring.
Dupont F–F 200 mm (40-way).	1	R31.00	Longer sensor runs.
M3 × 8 mm screws (pack of 10).	2	R6.00	Shell assembly.
M3 brass heat-set inserts (pack of 4).	2	R20.00	Strong threads in the printed shell (select 3 mm).
Heat-shrink assortment (127 pcs).	1	R34.00	Insulate joints.
NDE3D basket total (approx., excl. delivery)	~R305	All items in stock. Delivery free over R2000.	

Before ordering, verify your exact board: if your CYD is the USB-C revision (CYD2USB), the speaker header is standard. Confirm the 2-pin speaker connector is present on the edge near GPIO26 before relying on the on-board amp. If absent, you'll need a tiny PAM8403/MAX98357 amp later.

4. Wiring plan

VL53L0X ToF

VIN → 3V3 · GND → GND
 SDA → GPIO21 · SCL → GPIO22
 XSHUT → 3V3 (or leave unconnected)
 In code: `Wire.begin(21, 22)` — SDA first, then SCL.

Speaker

2-pin header → CYD SPK header (GPIO26 + GND)
 No external amp needed

Vibration motor (via transistor)

Motor + → 3V3 · Motor – → transistor collector
 1N4007 across motor (striped end → 3V3)
 Base → 1 kΩ → GPIO27 (PWM) · 10 kΩ base–GND
 Emitter → GND

Sound sensor

VCC → 3V3 · GND → GND
 OUT → GPIO35 (ADC / analog)

Pin budget (why it fits): the CYD only exposes GPIOs 21, 22, 27, 35. ToF = 21/22, motor = 27, sound = 35. — **exactly 4.** The old INMP441 needed 3 pins (SCK/WS/SD) which would not fit; the analog sound sensor needs only 1.

5. Build steps

1. **Flash/verify the CYD first:** Install Arduino core for ESP32, select “ESP32 Dev Module”, run one of the CYD example sketches (`TFT_eSPI` has a CYD user setup; `XPT2046_Touchscreen` for touch). Confirm screen + touch + USB-C work before anything else.
2. **Motor driver circuit on protoboard:** solder transistor, 1 kΩ + 10 kΩ resistors, 1N4007, and the two pin headers per the diagram above. Test with a blink sketch on GPIO27 at low PWM — the motor should buzz and the board should stay alive (if the board resets, the motor is drawing too much; drop to a smaller/proper pancake-motor current).
3. **Wire the sensors with Dupont leads:** VL53L0X to 3V3/GND/SDA/SCL (GPIO21/22, `Wire.begin(21,22)`); sound sensor OUT to GPIO35.
4. **3D-print the shell:** base/back for the CYD + speaker grille + ToF window + screw bosses. Install M3 brass inserts, mount the screen, seat the speaker behind its grille, and hot-glue the vibration motor to the inner shell wall or floor so the whole body shakes.
5. **Firmware (v1 pet logic):**
 - Face/eyes on screen: `TFT_eSPI` + `XPT2046_Touchscreen`
 - Proximity → `Adafruit_VL53L0X`
 - Clap/loud → sample GPIO35 ADC, trigger on a rising envelope threshold
 - Sounds → `ledcWriteTone` or DAC tones on GPIO26 (or .WAV via the amp)
 - Shiver → LEDC PWM burst on GPIO27 for ~300–600 ms when petted / happy
 - Mood state machine: idle → happy (pet) → curious (approached) → startled (clap)
6. **Assemble & route USB-C** out the back; fixate with the M3 + inserts.

6. Budget recap

LINE	AMOUNT	NOTES
NDE3D order (12 lines, incl. VAT)	~R305	Excl. delivery. Everything ships from one store — no second supplier.
Total electronics (est.)	~R305	Incl. VAT, excl. delivery. ~R100+ cheaper than the old two-store list.

7. Roadmap (future versions — not v1)

MODEL	ADD-ON	WHY IT HAPPENS LATER
v2 — camera	Move to an ESP32-S3 with OV2640 (e.g. XIAO ESP32S3 Sense or ESP32-S3-CAM)	The CYD has no camera interface and no spare pins. S3 adds PSRAM for vision models; enables person-following / face detection (e.g. ESP-WHO).
v3 — wheels	2× N20 gear motors + DRV8833 driver, Li-Po + TP4056 charger	Needs 4 more GPIOs + separate motor power; the S3 board swap in v2 makes room for it. Remove USB tether → battery.

8. Do NOT buy (for this version)

- Raspberry Pi Pico / micro:bit — your CYD is the controller.
- Battery / TP4056 / power switch — USB-C powers v1.
- Separate touch sensor (TTP223 etc.) — the CYD touchscreen does this.
- Audio amplifier module (PAM8403 / MAX98357) — the board has an on-board amp; only needed if you confirm your CYD lacks the speaker header.
- PNP-power stage is optional; the standard NPN + flyback diode covers v1's small motor.

Desk Pet v1 · generated from project build plan.